Marmik Shah

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EXPERIENCE

Loylty Rewardz - A BillDesk Group Company *Apr 2025 – Present*
Senior Software Developer *Mumbai, India*

- Architected a channel loyalty solution for manufacturers and finance companies, serving agents and channel partners with hierarchy and KPI management, leaderboards, tier computations, customer segmentation and approval workflows.
- Engineered a real-time, API-driven microservices system replacing T+1 bulk-file settlement, with 99% of transactions under 800ms; used Spring AOP for cross-cutting concerns and tracing, plus a webhook-based data-relay service.
- Delivered a low-latency balance service handling 300 TPS using caching, Bloom filters and database views, letting partners fetch balances and burn points across corporates and lines of business in a single call.
- Consolidated vendor integrations across multiple products into a unified hub using strategy-based orchestration and Kong gateway, lowering configuration and development effort by 60% when onboarding new vendors.
- Structured a bucket-based reward matrix framework and optimized enrichment and tenant loading, reducing database calls by over 45%, improving Hikari connection-pool utilization and cutting startup time from 300s to 85s.
- Shipped AI-assisted React admin panels and dev tools so client, product and relationship-manager teams can self-serve configuration changes without engineering support.
- Owned 7 loyalty programs end to end, handling BAU support and change requests as the primary point of contact for bank partners, while mentoring 3 developers through code reviews and design guidance.

Software Developer *Jun 2023 – Mar 2025*

- Implemented a Kafka-driven ingestion pipeline processing 1M+ records daily, combining polling and webhook patterns for asynchronous status tracking and reconciliation that catches missed status updates.
- Designed an enterprise-wide loyalty program that lets a single corporate span multiple lines of business through generic APIs.
- Developed a milestone-based journey rewards feature on GraphQL; architected a profile-driven multi-database setup and a strategy-driven post-award layer for cashback and gifting modules.
- Devised a parallel-processing lock for distributed ad hoc transactions and added budget controls to cap reward spend, with threshold-based actions and real-time updates.
- Achieved zero downtime on production services with keep-alive, virtual IPs and primary HA replicas; trimmed Redis memory by over 50% through a purging policy, a cluster setup and tuned RDB snapshots.
- Deployed microservices architecture on a Kubernetes cluster; introduced observability using ELK, Prometheus, and Grafana.

ComED Learning *Apr 2022 – Jul 2022*
Software Development Intern *Remote*

- Crafted UI/UX components and server-side functionality for the ComED Learning platform using Django and React.
- Connected backend APIs to enable data exchange and dynamic content rendering; resolved application bugs and ran functional tests.

KJ Somaiya College of Engineering *Feb 2021 – Apr 2021*
Web Development Intern *Mumbai, India*

- Created the website for Symphony, the college cultural fest, collaborating with the cultural and design teams.
- Styled mobile-friendly pages with interactive components and smooth animations that work across phones, tablets and desktops.

PROJECTS

Mumbai Police Portal — Full-stack complaint portal with station routing and status tracking (Django, PostgreSQL, Leaflet.js).

Newzer — Personalized news recommendation website that curates and summarizes articles based on users' interests, sentiments and feedback, with built-in fake news detection (Django, JavaScript, Machine Learning).

Waiterio Application — Client-server desktop app for restaurant ordering, connecting waiters and chefs (Java, Swing, Sockets).

EDUCATION

KJ Somaiya College of Engineering, *B.Tech in Computer Engineering | CGPI – 9.04* 2019 – 2023
Maharashtra State Board, *SSC – 89.80% | HSC – 84.62%* 2017 – 2019

TECHNICAL SKILLS

Programming Languages: Java, JavaScript, TypeScript, Python, SQL, C, C++, Solidity
Frameworks & Libraries: Spring, Spring Boot, Node.js, Django, React, Next.js, Vue.js
Databases & Caches: MySQL, PostgreSQL, Greenplum, SQL Server, MongoDB, Supabase, Redis, Neo4j
AI Tools: Claude, GPT, Cursor, Kiro, Copilot
DevOps & Infrastructure: CI/CD, Jenkins, Prometheus, Grafana, New Relic, ELK, Alertmanager, Linux, Bash, Nginx, Crontab
Tools & Platforms: Docker, Kubernetes, Git, Azure, Apache Camel, Airflow, Kong Gateway, Parcel
Messaging & APIs: Apache Kafka, RabbitMQ, REST, JPA, GraphQL, JWT, OAuth 2.0
Concepts: Microservices, Distributed Systems, Caching, High Availability, Bloom Filters, Blockchain, Multi-Tenancy

ACHIEVEMENTS

Awards: Rising Star (2023–24), Extraordinary Diligence (2024–25), Think Out of the Box (2025–26), Loylty Rewardz.
Hackathon: First place, HackerBash (CSI) hackathon, Team GrowMore, for the Co-Vaccine Web Portal (2021).